

<<Assignment Identifier>>

Lab 5.0

<<Report Title>>

Lab 5.0 - CouchDB

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NoSQL
(PROG3290)

Part II – CouchDB Challenge (For grading)

Company Alpha was recently acquired by Company Delta. Company Alpha kept didn't have very good database systems and kept all its files in excel spreadsheets.

Company Delta's database administrators would like to migrate everything into a single database as they use CouchDB as their central data repository.

You will need to assess the situation on Company Alpha, migrate the data into CouchDB and create the necessary views to easily navigate through the records.

1. Step 1

A. Convert the CSV File into JSON file.

- This can be found in the zip file you downloaded from eConestoga
- The recommended site to convert from CSV is

<https://products.aspose.app/cells/conversion/csv-to-json>

Note: you can use other sites for the conversion if you prefer.

2. Step 2

A. Create a CouchDB Database named Alpha and insert all the 15 records into the database.

- Remember that you will have to be attentive to the way you copy each of the 15 records into the database – you must pay attention on the JSON and remember to place a comma after the `_id`.
- Each record becomes a separate document within the database. **Do not copy all the 15 records into a single document as the query at the end will not work out.**

The screenshot displays the Apache CouchDB interface for a database named 'alpha'. The left sidebar contains navigation options: 'All Documents', 'Run A Query with Mango', 'Permissions', 'Changes', and 'Design Documents'. The main area shows a table view of 15 documents. Each document has a unique ID (e.g., 640cd5f934fc9f8314ba27f) and a corresponding JSON value. The table has columns for 'id', 'key', and 'value'. At the bottom, it indicates 'Showing document 1 - 15' and 'Documents per page: 20'.

id	key	value
640cd5f934fc9f8314ba27f	640cd5f934fc9f8314ba27f	{ "rev": "2-666f439653b7bf0f98177cf7194ec..." }
640cd5f934fc9f8314ba280	640cd5f934fc9f8314ba280	{ "rev": "1-1adeaee69bde93bc55d9ee0fcaa..." }
640cd5f934fc9f8314ba281	640cd5f934fc9f8314ba281	{ "rev": "1-7558d9ea03363e20ad3703045f1..." }
640cd5f934fc9f8314ba282	640cd5f934fc9f8314ba282	{ "rev": "1-27ec59bcad576027b8a53b6717b..." }
640cd5f934fc9f8314ba283	640cd5f934fc9f8314ba283	{ "rev": "2-0da563a8c5a6cb6e637a1bc3b8a..." }
640cd5f934fc9f8314ba284	640cd5f934fc9f8314ba284	{ "rev": "1-bbba2d3bcf9935cac150f93da867..." }
640cd5f934fc9f8314ba285	640cd5f934fc9f8314ba285	{ "rev": "1-clca1a41df06307797e6da8bc74f..." }
640cd5f934fc9f8314ba286	640cd5f934fc9f8314ba286	{ "rev": "1-2d870c131f0c9b44c8939aa5a9b..." }
640cd5f934fc9f8314ba287	640cd5f934fc9f8314ba287	{ "rev": "1-5e454f296c20f0072d52c3c6e27e..." }
640cd5f934fc9f8314ba288	640cd5f934fc9f8314ba288	{ "rev": "1-725fe77b64d51ac096ea6b4b66a..." }
640cd5f934fc9f8314ba289	640cd5f934fc9f8314ba289	{ "rev": "1-bf58a84a2630333e588ae83e47c..." }
640cd5f934fc9f8314ba28a	640cd5f934fc9f8314ba28a	{ "rev": "1-2b598b99f81d4fee4d354cef0531..." }
640cd5f934fc9f8314ba28b	640cd5f934fc9f8314ba28b	{ "rev": "1-e895b3669c101fe2897a366ed72..." }
640cd5f934fc9f8314ba28c	640cd5f934fc9f8314ba28c	{ "rev": "1-364090c1c096e62880139278fbb..." }
640cd5f934fc9f8314ba28d	640cd5f934fc9f8314ba28d	{ "rev": "1-e90ef3736554779e653ae620d6d..." }

3. Step 3

A. Create three views, as follows:

- View 1 – List by *original_title*.
 - Use Reduce function 'Count'
 - Once the view is generated, click on JSON on top-right of the screen and capture a screenshot. This is expected to return the total number of records along with other http information. **<Screenshot 1>**

The screenshot displays a web browser interface for configuring a new view in a database. The browser has several tabs open, including "Project Fauton - data", "JSON Conversion Issue", and "Microsoft Word - Laborat...". The address bar shows the URL "http://127.0.0.1:5984/_utils/#/database/alpha/new_view".

The interface is divided into a left sidebar and a main content area. The sidebar contains navigation options: "All Documents", "Run A Query with Mango", "Permissions", "Changes", and "Design Documents". The main content area is titled "New View" and includes the following fields:

- Design Document:** A dropdown menu set to "_design/" and a text input field containing "alpha_views".
- Index name:** A text input field containing "List_by_original_title".
- Map function:** A code editor containing the following JavaScript function:

```
1 function(doc) {  
2   if (doc.original_title) {  
3     emit(doc.original_title, 1);  
4   }  
5 }
```
- Reduce (optional):** A dropdown menu set to "_count".

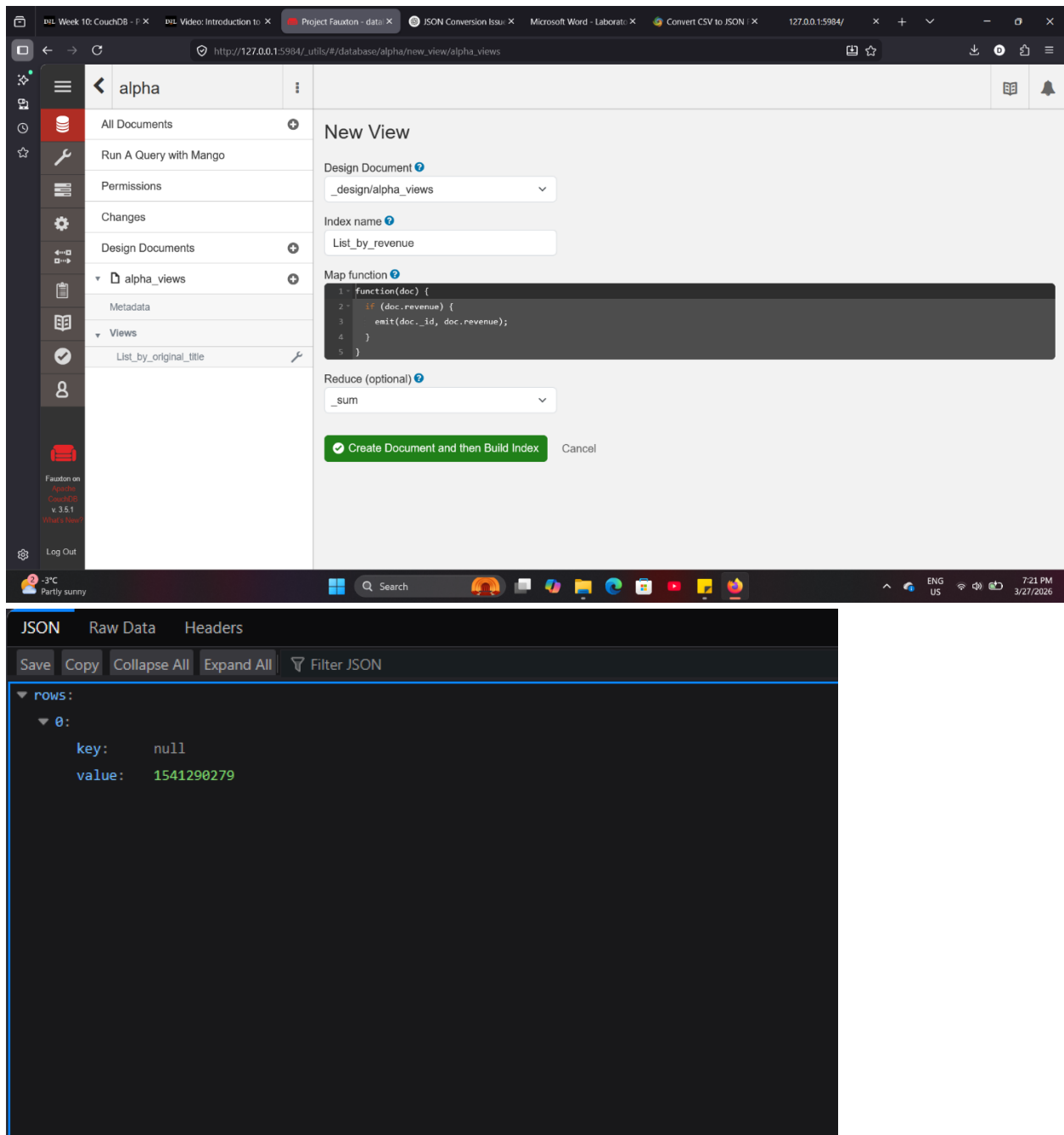
At the bottom of the "New View" form, there are two buttons: "Create Document and then Build Index" (highlighted in green) and "Cancel".

Below the browser window, a JSON viewer is shown. It has tabs for "JSON", "Raw Data", and "Headers", and buttons for "Save", "Copy", "Collapse All", "Expand All", and "Filter JSON". The JSON viewer displays the following data:

```
POWS:  
0:  
  key: null  
  value: 15
```

b. View 2 – List by *revenue*.

- Use Reduce function 'Sum'
- Once the view is generated, click on JSON on the top-right of the screen and capture the screenshot of the results. This is expected to return the sum of revenue along with other http information. <Screenshot 2>



c. View 3 – List by *revenue*.

- Use Reduce function 'Stats'
- Once the view is generated, click on JSON on the top-right of the screen and capture the screenshot of the results. This is expected to return relevant revenue statistics along with other http information.

<Screenshot 3>

The screenshot displays the CouchDB 'Edit View' interface for a database named 'alpha'. The left sidebar shows a tree view with 'alpha_views' expanded, listing 'List_by_revenue_stats' as the selected view. The main area shows the 'Edit View' configuration: 'Design Document' is '_design/alpha_views', 'Index name' is 'List_by_revenue_stats', and the 'Map function' is a JavaScript function that emits document IDs and revenue. The 'Reduce (optional)' dropdown is set to '_stats'. A green button 'Save Document and then Build Index' is visible. Below the browser, a JSON viewer shows the result of the reduce function: an object with 'sum', 'count', 'min', 'max', and 'sumsq' values.

```
function(doc) {  
  if (doc.revenue) {  
    emit(doc._id, doc.revenue);  
  }  
}
```

```
{  
  "sum": 1541290279,  
  "count": 11,  
  "min": 10017322,  
  "max": 373554033,  
  "sumsq": 396472243362804769  
}
```

4. Step 4

- A. Create a new database in CouchDB with a name Master
- B. Using the Replication function, replicate all the data from Alpha database into Master. Take a screenshot of the configuration you applied to the configuration screen. This is simply the data you inserted into the Replication screen. **<Screenshot 4>**

Job Configuration

Source

Type: Local database

Name: alpha

Authentication: Username and password

Username: porewen3848_admin

Password: [masked]

Target

Type: Existing local database

Name: master

Authentication: Username and password

Username: porewen3848_admin

Password: [masked]

Options

Replication type: One time

Replication document: Custom ID (optional)

Clear Start Replication

5. Step 5

- A. Check to see if the data has been replicated into the Master Database by doing the following:
 - a. Collect a screenshot of the replication screen to demonstrate the replication happened.

<Screenshot 5>

Replication

Polling Interval: 5 minutes Refresh

Replicator DB Activity: _replicate Activity

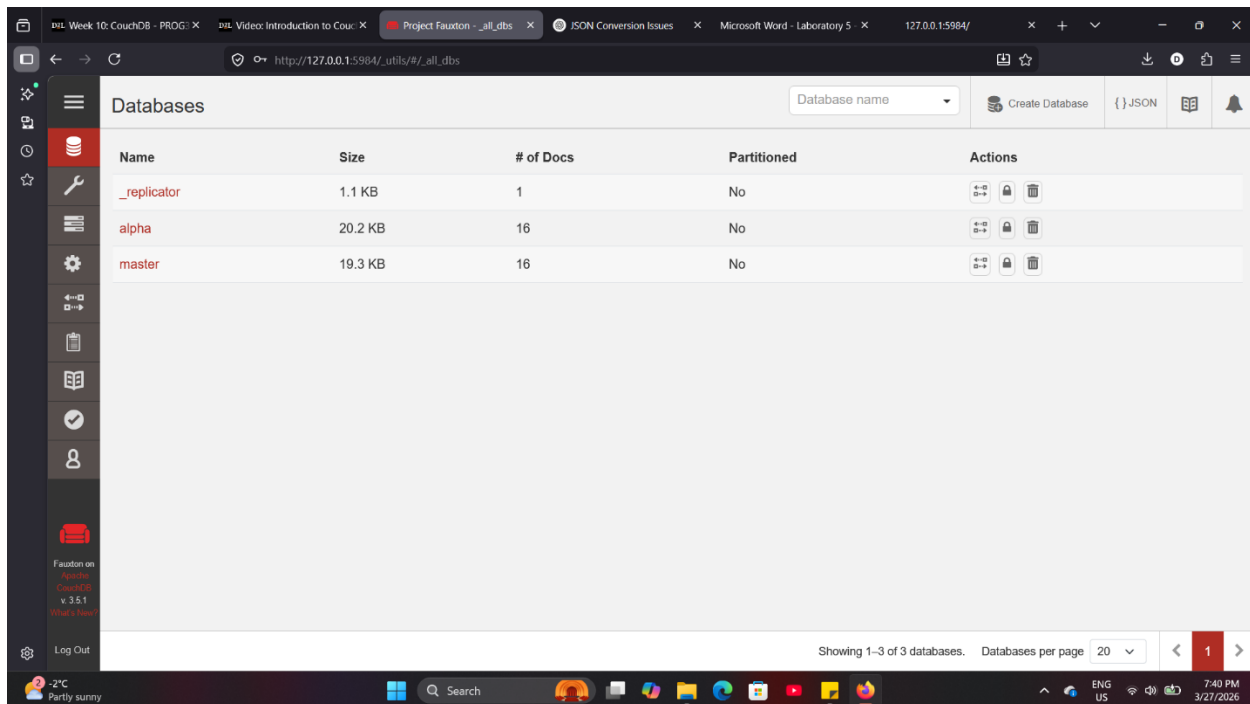
Replications must have a replication document to display in the following table.

Filter replications

Source	Target	Start Time	Type	State	Actions
http://127.0.0.1:5984/alpha	http://127.0.0.1:5984/master	Mar 27th, 7:38 pm	One time	Completed	[Icons]

Showing replications 1 - 1 Max replications displayed: 100

- b. Check that the data has been successfully replicated by opening the Master database and manually checking that it is the same data you had on the initial database.
- c. Take a screenshot of CouchDB summary screen that shows that the number of records is the same in both databases <Screenshot 6>



6. Step 6

- In your opinion, how should Company Delta plans to maintain the continual flow of data from Company Alpha?
Company Delta should implement continuous replication from Company Alpha's data sources to the Master CouchDB database.
- Based on what you have learned in this course so far, what other database tools can Company Delta use to transfer all the data from Company Alpha? Why?
Database Migration Services -AWS Database Migration Service (DMS)
They are reliable and industry standard.
- Are these services available through a Database as a Service Provider? If your answer is yes, please provide examples.
Yes, many DB as a Service providers offer tools and services for data migration and replication.
Amazon AWS DynamoDB
Azure Cosmos DB
